KAIZEN MATHS

Lesson starter

Integration of Algebraic Fractions

Logarithmic Forms

Name _____

S1

Integrate each algebraic fraction by using the linear denominator logarithm form.

Find $\int \frac{4}{4x-6}$ dx.

S2

Identify the derivative of the denominator before integrating.

Find $\int \frac{2x-6}{x^2-6x+7}$ dx.

S3

Rewrite the algebraic fraction as a constant plus a proper fraction, then integrate.

Find $\int \frac{-12x-8}{3x+3}$ dx.

S4

Integrate each algebraic fraction by using the linear denominator logarithm form.

Find $\int \frac{15}{5x+3}$ dx.

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Integration of Algebraic Fractions

Logarithmic Forms

Name

E1

Identify the derivative of the denominator before integrating.

Find $\int \frac{2x+2}{x^2+2x+12}\,dx$.

E2

Rewrite the algebraic fraction as a constant plus a proper fraction, then integrate.

Find $\int \frac{-4x+7}{4x-5}\,dx$.

E3

Integrate each algebraic fraction by using the linear denominator logarithm form.

Find $\int \frac{-16}{4x-1}\,dx$.

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Find $\int \frac{-16}{4x-1}\,dx$.

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Teacher answers & working

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S1

Integrate each algebraic fraction by using the linear denominator logarithm form.

Find $\int \frac{4}{4x-6}\,dx$.

Answer:

$\ln|4x - 6| + C$

Working:

Let $u = 4x - 6$.

$\frac{du}{dx} = 4$, so $dx = \frac{du}{4}$.

$\int \frac{4}{4x-6}\,dx = \int \frac{4}{u}\,\frac{du}{4}$.

$= 1\int \frac{1}{u}\,du$.

$= \ln|u| + C$.

Substitute u = 4x − 6: ln|4x − 6| + C.

S2

Identify the derivative of the denominator before integrating.

Find $\int \frac{2x-6}{x^2-6x+7}\,dx$.

Answer:

$\ln|x^2 - 6x + 7| + C$

Working:

Let $f(x) = x^2 - 6x + 7$.

Then f'(x) = 2x − 6.

The numerator is exactly the derivative of the denominator.

$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$.

Therefore $\int \frac{2x-6}{x^2-6x+7}\,dx$ = ln|x² − 6x + 7| + C.

S3

Rewrite the algebraic fraction as a constant plus a proper fraction, then integrate.

Find $\int \frac{-12x-8}{3x+3}\,dx$.

Answer:

$-4x + \frac{4}{3}\ln|3x + 3| + C$

Working:

Rewrite the numerator in terms of the denominator.

$-12x - 8 = -4(3x + 3) + 4$.

So $\frac{-12x-8}{3x+3} = -4 + \frac{4}{3x+3}$.

$\int \frac{-12x-8}{3x+3}\,dx = \int -4\,dx + \int \frac{4}{3x+3}\,dx$.

$= -4x + \frac{4}{3}\ln|3x + 3| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Logarithmic Forms

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S4

Integrate each algebraic fraction by using the linear denominator logarithm form.

Find $\int \frac{15}{5x+3}\,dx$.

Answer:

$3\ln|5x + 3| + C$

Working:

Let $u = 5x + 3$.

$\frac{du}{dx} = 5$, so $dx = \frac{du}{5}$.

$\int \frac{15}{5x+3}\,dx = \int \frac{15}{u}\,\frac{du}{5}$.

$= 3\int \frac{1}{u}\,du$.

$= 3\ln|u| + C$.

Substitute u = 5x + 3: $3\ln|5x + 3| + C$.

E1

Identify the derivative of the denominator before integrating.

Find $\int \frac{2x+2}{x^2+2x+12}\,dx$.

Answer:

$\ln|x^2 + 2x + 12| + C$

Working:

Let $f(x) = x^2 + 2x + 12$.

Then $f'(x) = 2x + 2$.

The numerator is exactly the derivative of the denominator.

$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$.

Therefore $\int \frac{2x+2}{x^2+2x+12}\,dx = \ln|x^2 + 2x + 12| + C$.

E2

Rewrite the algebraic fraction as a constant plus a proper fraction, then integrate.

Find $\int \frac{-4x+7}{4x-5}\,dx$.

Answer:

$-x + \frac{1}{2}\ln|4x - 5| + C$

Working:

Rewrite the numerator in terms of the denominator.

$-4x + 7 = -1(4x - 5) + 2$.

So $\frac{-4x+7}{4x-5} = -1 + \frac{2}{4x-5}$.

$\int \frac{-4x+7}{4x-5}\,dx = \int -1\,dx + \int \frac{2}{4x-5}\,dx$.

$= -x + \frac{1}{2}\ln|4x - 5| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Logarithmic Forms

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E3

Integrate each algebraic fraction by using the linear denominator logarithm form.

Find $\int \frac{-16}{4x-1}\,dx$.

Answer:

$-4\ln|4x-1|+C$

Working:

Let $u = 4x - 1$.

$\frac{du}{dx} = 4$, so $dx = \frac{du}{4}$.

$\int \frac{-16}{4x-1}\,dx = \int \frac{-16}{u}\,\frac{du}{4}$.

$= -4\int \frac{1}{u}\,du$.

$= -4\ln|u| + C$.

Substitute u = 4x − 1: $-4\ln|4x-1| + C$.